SIMATS SCHOOL OF ENGINEERING
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CHENNAI-602105

Engineer to Excel

TECHNICAL QUESTIONS FOR UST GLOBAL – SET3

1. Write a program to print the following output for the given input. You can assume the string is of odd length

Eg 1: Input: 12345

Output:

```
1   5
 2 4
  3
 2 4
1   5
```

2. Given two strings A and B, find if A is a subsequence of B.

Example 1:

Input:

A = AXY

B = YADXCP

Output: 0

Explanation: A is not a subsequence of B

as 'Y' appears before 'A'.

Example 2:

Input:

A = gksrek

B = geeksforgeeks

Output: 1

Explanation: A is a subsequence of B.

3. Given two sorted arrays arr1[] and arr2[] of sizes n and m in non-decreasing order. Merge them in sorted order without using any extra space. Modify arr1 so that it contains the first N elements and modify arr2 so that it contains the last M elements.

Example 1:

Input:

n = 4, arr1[] = [1 3 5 7]

m = 5, arr2[] = [0 2 6 8 9]

Output:

arr1[] = [0 1 2 3]

arr2[] = [5 6 7 8 9]

Explanation:

After merging the two

non-decreasing arrays, we get,

0 1 2 3 5 6 7 8 9.

Example 2:

Input:

n = 2, arr1[] = [10 12]

m = 3, arr2[] = [5 18 20]

Output:

arr1[] = [5 10]

arr2[] = [12 18 20]

Explanation:

After merging two sorted arrays

we get 5 10 12 18 20.

4. Given a String S, reverse the string without reversing its individual words. Words are separated by dots.

Example 1:

Input:

S = i.like.this.program.very.much

Output: much.very.program.this.like.i

Explanation: After reversing the whole

string(not individual words), the input

string becomes

much.very.program.this.like.i

Example 2:

Input:

S = pqr.mno

Output: mno.pqr

Explanation: After reversing the whole

string , the input string becomes

mno.pqr

5. Given a 2-dimensional matrix of size ‘N’ x ‘M’, rotate the elements of the matrix clockwise.

For example:

Input Matrix: [[1, 2, 3]

[4, 5, 6]

[7, 8, 9]]

Output Matrix: [[4, 1, 2]

[7, 5, 3]

[8, 9, 6]]

The output matrix is generated by rotating the elements of the input matrix in a clockwise direction. Note that every element is rotated only once.

Note :

You do not need to print anything; it has already been taken care of. Also, update the given matrix in-place.

Detailed explanation (Input/output format, Notes, Images)

Constraints :

1 <= T <= 10

1 <= N, M <= 100

-10^5 <= data <= 10^5,

where ‘T’ is the number of test cases, ‘N’ and ‘M’ are the numbers of rows and columns respectively and ‘data’ is the value of the elements of the matrix.

Sample Input 1 :

1

4 4

1 2 3 4

5 6 7 8

9 10 11 12

13 14 15 16

Sample Output 1 :

5 1 2 3

9 10 6 4

13 11 7 8

14 15 16 12

6.You are given a 'N' * 'N' maze with a rat placed at 'MAZE[0][0]'. Find and print all paths that rat can follow to reach its destination i.e. 'MAZE['N' - 1]['N' - 1]'. Rat can move in any direc-tion (left, right, up and down).

Value of every cell in the 'MAZE' can either be 0 or 1. Cells with value 0 are blocked means the rat can-not enter into those cells and those with value 1 are open.

Detailed explanation (Input/output format, Notes, Images)

Constraints:

1 <= N <= 10

0 <= MAZE[i][j] <=1

Where 'MAZE[i][j]' denotes the value in the cell of 'MAZE'.

Time Limit: 1 sec

Sample Input 1 :

3

1 0 1

1 0 1

1 1 1

Sample Output 1 :

1 0 0 1 0 0 1 1 1